

The 14th International Symposium on Equatorial Aeronomy (ISEA) Report



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The *International Symposium on Equatorial Aeronomy (ISEA)* is held approximately every three to four years, and is a major gathering of scientists around the world interested in studying the low-latitude atmosphere and ionosphere, as well as coupling with higher latitudinal regions. Each ISEA represents an opportunity for researchers to share their most recent results as well as discuss possibilities for future campaigns and experiments.

Despite the fact that much progress has been made in the study of ionospheric density structure and dynamics in the last decade, there are many gaps remain in the global understanding of the fundamental electrodynamics that governs the formation of equatorial ionospheric density irregularities. This is in part due to the uneven distribution of ground-based instruments, largely due to the lack of ground-based instrumentation in Africa. Until recently, ionospheric density structure in the African sector has been only estimated by model interpolation over vast geographic areas. However, satellite data have shown that the equatorial ionospheric density structures in the African continent, respond to space weather effects differently than do other parts of the Earth. However, these unique responses have not been confirmed, validated, or studied in detail due to the lack of suitable ground-based instrumentation in the region. Thus, the cause of these unique density irregularities in the continent, including their temporal and spatial evolutions, remains a mystery for the scientific community. Therefore, to gain a better understanding, almost a decade ago the scientific communities, under the umbrella of United Nations-sponsored International Space Weather Initiative (ISWI) the then International Heliophysical Year (IHY), showed significant interest to close the largest land-based gap in ground-based instruments on the continent which was devoid of ground-based instruments (see left panel in [Figure 1](#)). Because, a decade ago when we look at a map of the world that shows the location of ground-based space physics instrumentations (radars, magnetometers, ionosondes, GPS dual frequency receivers, and others), one can quickly recognize Africa's lack of space physics research infrastructure. Thus during the last few years, quite significant number of small-to-medium scale instruments have been either deployed or in the process of deployment in Africa as shown in [Figure 1](#) (right panel). In addition to new scientific discoveries and advancing space science research in Africa by establishing scientific collaborations between scientists in developed and developing nations, the deployment of those instruments spontaneously increase the number of African space scientists as well as boost the importance of space science awareness in many African countries.

The scientific communities also recognize that in order to develop space science research infrastructure within Africa, space science educational infrastructure also needs to be developed to support the long-term operation and use of the science instrumentation. One way to address this concern is to increase and facilitate a continuous strong interaction between scientists from developed countries and African young professors and postgraduate students. In order to create such interaction between the two parties, the space science communities organize idealistic and scientific meetings and held in Africa. In response to these needs, several international workshops/conferences and summer schools were already conducted in different part of Africa. The very successful IHY-Africa 2007 (held in Addis Ababa, Ethiopia in November 2007), IHY-Africa 2009 workshop (held in Livingstone, Zambia in June 2009), IHY summer school (held in Abuja, Nigeria in 2009), ISWI summer school (held in Bahir Dar, Ethiopia in 2010), and International AGU Chapman Conference (held in Addis Ababa, Ethiopia in 2012) are among

those that had been held in Africa. More recently, in recognition to the unprecedented achievement of the space science development in Africa, the international scientific community decided and held one of its prestigious Space Science meetings, The 14th International Symposium on Equatorial Aeronomy (ISEA), in Africa for the first time in more than four decades. In general, such meetings held in Africa have opened the door for many new collaborations between western and African institutions to study the complex low- and mid-latitude space weather systems in great detail.

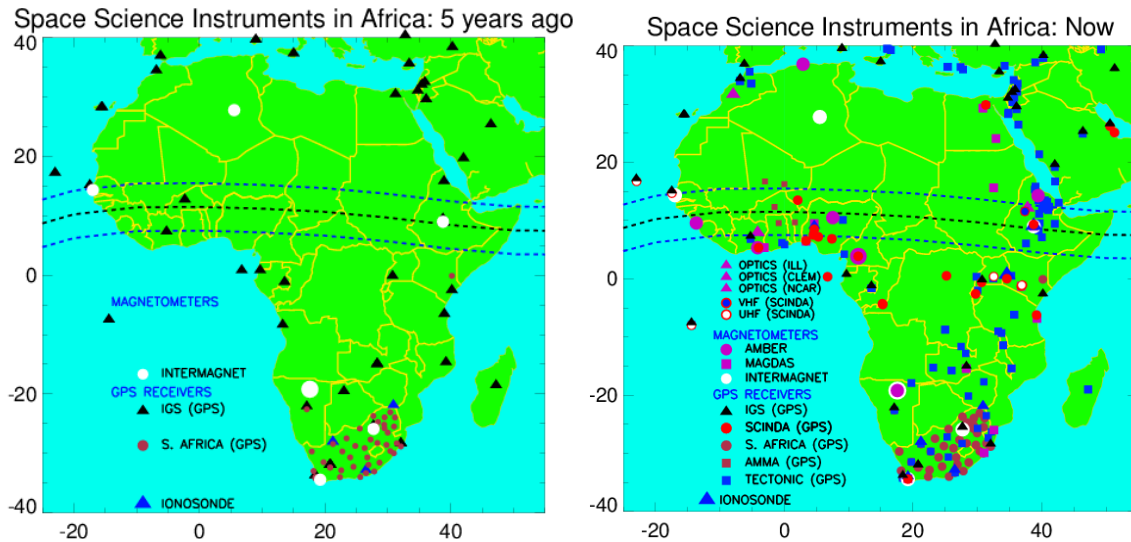


Figure 1. (left panel) Ground-based instrument coverage in African five years ago, and (right panel) shows the current ground-based instrumentation in Africa. Yellow big circle is the proposed site.

Attendees and Logistics

The 14th ISEA conference has been held successfully in Bahir Dar, Ethiopia during 19-23 October 2015. The conference became possible through two committees; the International Main Scientific Organizing (MSO) Committee, and the Local Organizing Committee (LOC). A complete list of members of these committees appears in Appendix A. The conference has been well attended by excess of 125 participants (see Figure 2), represented from 27 nations (see Figure 3). Of these delegates about 45% represented from more than 10 different African countries, about 25% were from USA, and the remaining 30% were from other countries that include Europe, Asia, and South America. Because of the generosity of the sponsors (see Appendix E), about 61 African delegates and 27 delegates from outside African were sponsored either partially or in full. Travel arrangements for the selected sponsored African scientists were made using a travel agency in the USA, Austria, and Japan coordinated by Boston College, UNOOSA, MIT, and Nagoya University. The travel arrangements for the sponsored US scientists and other delegates outside African and USA were also made in the US through Boston College, MIT, and DLR in Germany. The conference participants, including those who received full or partial sponsorship, were accommodated in two hotels (Grand Resort and Spa and

Jacaranda Hotels in Bahir Dar). The LOC in collaboration with the MSO coordinated all the accommodation arrangements with the hotels. *Figure 3* shows the geographic distribution of countries around the world represented at the 14th ISEA conference in Ethiopia. The list of countries represented at the workshop is given in Appendix C, and a complete list of delegates can be found in Appendix D.



Figure 2: Group photo of the participants take outside the meeting hall, behind Lake Tana (the largest lake in Africa and is the source of Blue Nile).

The conference was held at the Grand Resort and Spa Bahir Dar conference hall. At the opening ceremony of the conference several presentations were made by different agency representatives, including Dr J. Meriwether (NFS's representative), Dr R. Pfaff (NASA's representative), Dr. K. Shiwakowa (Japan's funding representative), Dr. R. Caton (AFOSR's representative), and others.

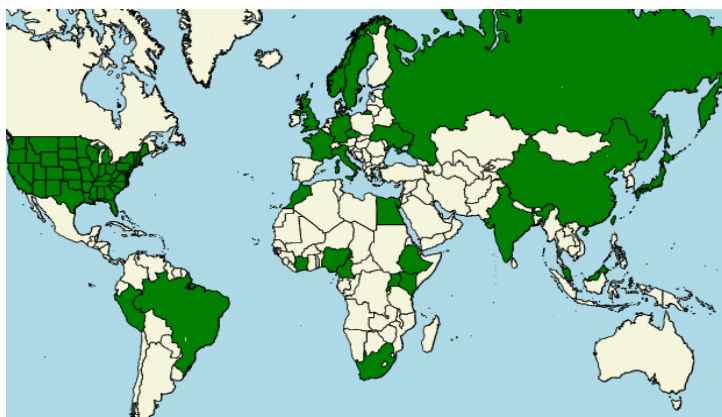


Figure 3: Geographic distribution of countries (shaded in green) that were represented at the 14th ISEA Conference in Bahir Dar, Ethiopia

Scientific Program

The conference has been organized for two prime objectives: (1) to assemble an international group of heliophysics scientists to plan and discuss current and needed observations at mid and low latitudes in the African longitude sector, a region that has never been explored in detail using ground-based instruments, and (2) to enhance the space science education and research interest in the Africa continent. The first half day (Monday morning) were dedicated to the opening ceremony followed by presentation by three key note addresses about; Achievements in Space Science: Capacity building GIRGEA: 1991-2015 (by Prof C. Amory-Mazaudier of France); Education and Outreach Programs, and Opportunities and Success Stories for the Equatorial Regions (by Prof P. Doherty of USA); ICTP efforts for the advancement of Aeronomy and Space Weather related science in Africa (by Prof S. Radicella of Italy). The status of the African Geophysical Society (AGS), which was officially inaugurated at the closing ceremony of Chapman Conference in 2012, has been also presented at the opening ceremony by its current president, Prof B. Rabiou of Nigeria.

In addition to the panel discussion sessions at the end of the conference, the remaining conference program included six main science themes:

1. *Magnetosphere-Ionosphere-Thermosphere coupling impact at low- and mid-latitude*
2. *Longitudinal dependence of equatorial electrodynamics and storm-enhanced density*
3. *Ionospheric irregularities and scintillations*
4. *New results from recently deployed instrumentation*
5. *Equatorial aeronomy related to atmosphere-ionosphere coupling*
6. *Future opportunities using upcoming new mission and planned ground-based instrumentation*

Very compelling and high standard oral and poster papers, for each scientific theme, were presented at the conference. For all five science sessions as well as for *Future opportunities using upcoming new mission and planned ground-based instrumentation* session there were about 76 oral presentations, with the main focus on observation and modeling efforts of understanding the unique space weather events in the African region. In addition, the conference had one poster sessions as well as one panel discussion on Infrastructure and Capacity Building. The poster sessions, which included about 40 poster presentations, were divided according to the main science sessions mentioned above. The panel discussion, which includes seven panelist (Dr. T. Immel (USA), Dr. N. Jakowski (Germany), Prof. J. Makela (USA), Prof. S. Radicella (Italy), and Prof. B. Rabiou (Nigeria)) and chaired by Dr. P. Erickson, was designed to find a way forward to address the most important question not only for African scientist but also for the international scientific communities in general, i.e., what type of local and international support are needed to increase the ground-based instruments deployment in the region to address the outstanding science questions, such as why African sector ionospheric irregularities are so unique and active all year round compared to other longitudinal sectors?

Conclusion Remarks: The 14th ISEA conference was successful in all aspect. In additions being a venue of scientific brainstorm discussion, the meeting served as a vehicle and brought African Space Scientists together and providing a forum in which they could meet and make contacts with international colleagues in the field. A number of collaborations have been established as a result of such similar conferences and workshops. More importantly, quite a number of students were exposed to the Space Science field and had opportunity to have a one-on-one discussion with successful scientists in the field (see Figure 4 & 5 below). It is very imperative that

meetings like the 14th ISEA conference can form a very valuable part of scientific research not only for Africans but also for scientists from different part of the world, and should be highly encouraged and supported.

The 14th ISEA conference would not have been possible without the support of the sponsors, like NSF, NASA, AFOSR, ICTP, SCOSTEP, UNOOSA, EGU, Boston College, JSPS, and Nagoya University, and the Ethiopian government and other local sponsors (Bahir Dar University, and Entoto Observatory and Research center). The complete list of sponsors is shown in Appendix E. The MSOs are also thankful for the support and hard work of local organizing team.



Figure 4: African professor discussing his results with scientist from USA (left panel) and an African student having private discussion with scientist from Europe about his research work (right panel).



Figure 5: Left panel shows participants from different region socializing and discussing their result each other, and right panel depicts an African professor discussing his poster presentation with scientist from USA about his research work.

Appendix A

List of Conveners and Organizing Committees

Main Scientific Organizing Committee

Endawoke Yizengaw, Boston College, USA

Jonathan Makela, University of Illinois at Urbana-Champaign, USA

Jorge L. Chau, Leibniz Inst. of Atmos. Phys. at the Rostock University, Germany

Philip Erickson, MIT Haystack Observatory, USA

Kazuo Shiokawa, Nagoya University, Japan

Duggirala Pallam Raju, Physical Research Laboratory, India

Claudia Stolle, Helmholtz Centre Potsdam GFZ, Germany

Anja Stromme, SRI International, USA/Norway

Marco Milla, Jicamarca Radio Observatory, Peru

Local Organizing Committee (LOC) – Ethiopia

Baylie Damtie, Bahir Dar University (LOC Chairman)

Melessew Nigussie, Bahir Dar University

Abyiot Bires, Bahir Dar University

Tsegaye Kassa, Bahir Dar University

Ambelu Tebabal, Bahir Dar University

Fasil Tesema, Bahir Dar University

Moges Wasie, Bahir Dar University

Esubalew Alemneh, Bahir Dar University

Oriyon Mengistu, Bahir Dar University

Nigist Getachew, Bahir Dar University

Selam Andualem, Bahir Dar University

Appendix B

Conference Program

Time	Title	Presenter	Convener
Sunday (18 October 2015)			
1700 - 2100	Registration followed by welcome reception		
Monday (19 October 2015)			
Opening Ceremony			
0830-0925	Late registration		
0925-0945	Welcome MSO/LOC		
0945-1000	NSF Rep. (J. Meriwether)	J. Meriwether	
1000-1015	NASA Rep. (E. Talaat)	E. Talaat (R. Pfaff)	
1015-1030	JAPAN	K. Shiokawa	
1030-1100	Coffee Break		
1100-1120	Achievements in Space Science: Capacity building GIRGEA: 1991-2015. (<i>Keynote</i>)	C. Amory-Mazaudier	
1120-1140	Education and Outreach Programs, and Opportunities and Success Stories for the Equatorial Regions. (<i>Keynote</i>)	P. Doherty	
1140-1200	ICTP efforts for the advancement of Aeronomy and Space Weather related science in Africa. (<i>Keynote</i>)	S. M. Radicella	
1200-1215	AGS	B. Rabiou	

Damtie

1215-1330	Lunch Break		
Session 1			
1330-1350	Stormtime Impact on the Low- to Mid-Latitude Ionosphere. (<i>Invited</i>)	J. Huba	Goncharenko & Yamazaki
1350-1405	Magnetosphere-Ionosphere Coupling Effects at Mid Latitudes	J. C. Foster	
1405-1420	Low latitude ionospheric response to the occurrence of High Speed Streams	C. M. N. Candido	
1420-1435	Simulation of the low latitude ionospheric response to an intense space weather event.	L. M. Joshi	
1435-1450	Low latitude ionospheric dynamics as a possible source of poleward traveling ionospheric disturbances during geomagnetic storms.	J. B. Habarulema	
1450-1510	C/NOFS Observations of Reversed Zonal E x B Drifts Below the Equatorial Ionospheric F-Peak at Sunset and Their Implications for the Generation of Large Scale Instabilities. (<i>Invited</i>)	R. Pfaff	Goncharenko & Yamazaki
1510-1525	Ionospheric response to St. Patrick's day geomagnetic storm of 2015	B. Nava	
1525-1540	GSM TIP simulation of the plasmasphere and low-latitude and equatorial ionosphere F region disturbances during	M. Klimenko	
1540-1605	Coffee Break		
1605-1625	Effects of Storm-time Neutral Winds and Penetration Electric Fields on the Mid- and Low-Latitude Ionosphere and Thermosphere. (<i>Invited</i>)	G. Lu	Stolle & Habarulema
1625-1645	Day-to-day variations in the solar quiet (Sq) current	Y. Yamazaki	

	system. (<i>Invited</i>)		
1645-1700	Current circuit connecting the polar and equatorial region deduced by a global ionospheric potential solver	A. Nakamizo	
1700-1715	Time-dependent generalized Ohm's Law and formation of global Cowling channel in the ionosphere	A. Yoshikawa	
1715-1730	Inner Magnetospheric Influences on Low and Mid Latitude Ionospheric Variability	P. J. Erickson	
1730-1745	A Probe of Magnetosphere-Ionosphere Coupling using the Propagation Characteristics of Very Low Frequency Signal	V. U. J. Nwankwo	
1745-1800	Modeling Plasma drift velocities within the African sector	M. Dubazane	
Tuesday (20 October 2015)			
Session 2			
0800-0820	Possible sources of longitudinal differences in equatorial spread F detection. (<i>Invited</i>)	D. Hysell	Erickson
0820-0840	Longitudinal Variation Of Equatorial Electrojet And The Occurrence Of Its Counter Electrojet. (<i>Invited</i>)	A. B. Rabiou	
0840-0855	On the longitudinal dependence of the equatorial electrojet	V. Doumbia	
0855-0910	Observations Of Equatorial Ionization Anomaly Over Africa During A Year Of Deep Minimum	O. S. Bolaji	
0910-0925	Longitudinal and hemispheric Effects on Equatorial Spread F Occurrence	J. O. Adeniyi	
0925-0940	The Correlation between the Longitudinal Dependence of Equatorial Irregularities and	E. Yizengaw	

	Electrodynamics		
0940-1000	Geomagnetic Storm Effects in Total Electron Content during Solar Cycle 24. (<i>Invited</i>)	A. Coster	
1000-1030	Coffee Break		
1030-1050	Longitudinal Dependence of Equatorial Ionospheric Effects Based on An Event-Driven Multi-GNSS Data Collection System. (<i>Invited</i>)	J. Morton	
1050-1105	Observation of the mid-and low-latitude ionosphere: SSUI and GUVI data	L. J. Paxton	
1105-1120	Storm Enhanced Density Total Electron Content Base and Plume Properties Over North America as a Function of Storm Phase.	M. Moldwin	
1120-1135	Daytime EEJ current strength and vertical plasma drifts: The conjugate point	J. Adeniyi	
1135-1150	Longitudinal dependence of thermospheric neutral winds: Results from ground-based FPI observations	J. Makela	
1150-1205	New Results on the Midnight Temperature Maximum for Middle and Equatorial Latitudes	R. Mesquita	
1205-1345	Lunch Break		
Session 4			
1345-1400	The Low Latitude Ionospheric Sensor Network - Recent Scientific Results	C. E. Valladares	
1400-1415	Radar and satellite studies of equatorial spread F in the Brazilian sector	F. da S. Rodrigues	
1415-1435	The Blue Nile Radar (BNR): Ionospheric Monitoring from the first coherent backscatter radar in Africa.	R. G. Caton	

Hysell & Doumbia

Meriwether & Mahrous

	(Invited)		
1435-1450	A study of the African equatorial ionospheric irregularity using Coherent Backscatter Radar, C/NOFS satellite, and GPS receiver data	M. Nigussie	
1450-1505	Investigations on the wave dynamics of daytime upper atmosphere over low-latitudes	D. Pallamraju	
1505-1520	Equatorial electrojet over West Africa from ground and Swarm satellites	P. Coisson	
1520-1535	An HF beacon network for monitoring ionospheric irregularities at low magnetic latitudes	D. Hysell	
1535-1620	Coffee Break		
			Rabiu & Valladares
1620-1635	Equatorial ionization anomaly probed by FORMOSAT-3/COSMIC	J-Y. Liu	
1635-1650	Ionospheric Response to Magnetar Flare: Detection of SGR J1550-5418 by Coherent Ionospheric Doppler Radar	A. Mahrous	
1650-1710	Networked Fabry-Perot measurements of the 630.0-nm emission. (Invited)	B. Harding	
1710-1725	New Results on Optical Measurements of Ion Temperatures and F-region Plasma Drifts	J. Meriwether	
1725-1740	Wind field estimations in the mesosphere and lower thermosphere over northern Germany with MMARIA: Comparisons of different approaches	J. L. Chau	
1740-1755	Geomagnetic observatory data from Huancayo, Peru, to investigate long-term trends in the equatorial electrojet region	J. Matzka	
1755-1810	Results from the esa-alcantara equatorial ionosphere	M. Orue	

	over Africa (IONAF)		
1930-2200	Conference dinner		
Wednesday (21 October 2015)			
Excursion			
	08:30 - 16:30 Excursion		
Poster Session			
	18:00 - 21:00 Poster Session		
Thursday (22 October 2015)			
Session 3			
0800-0820	Equatorial F region irregularities observed by GIRI. (<i>Invited</i>)	A. K. Patra	Rodrigues & Adeniyi
0820-0835	Role of thermospheric meridional winds in controlling the Equatorial Spread F (ESF) irregularities: A comparative study using h'F and hpF2 data	S. Sreekumar	
0835-0850	Three-dimensional high-resolution plasma bubble modeling and its integration with the whole atmosphere-ionosphere coupled model	T. Yokoyama	
0850-0905	C/NOFS Observations of Combined Electric Field, Magnetic Field, and Plasma Density Observations of Plasma Structures and Instabilities in the Low Latitude Ionosphere	R. Pfaff	
0905-0920	The progression towards electrostatic equilibrium in E-F coupling, & whether it ever arrives	R. Cosgrove	
0920-0935	High Frequency Equatorial Electrojet Fluctuation inferred from model	E. Mengistu	

0935-0950	Equatorward displacements of OI 630 nm airglow intensity depletions imaged at East Asia Sector	F. C. de Meneses Jr	
0950-1010	Imaging observation of the low- and mid-latitude ionosphere from ground and space. <i>(Invited)</i>	A. Saito	
1010-1035	Coffee Break		
1035-1050	Daytime zonal drifts in the ionospheric E and 150 km regions estimated using EAR observations	P. P. Chaitanya	Makela & V. Narayanan
1050-1105	Swarm High Resolution Density Measurements and GPS Track Losses	S. Buchert	
1105-1125	Scintillation measurements at Bahir Dar during the high solar activity phase of solar cycle 24. <i>(Invited)</i>	N. Jakowski	
1125-1140	Climatology of equatorial electron-density perturbations observed by DEMETER satellite (2005-2010) at night time	T. Nguyen	
1140-1155	The ionospheric GPS scintillations observed by using radio occultation observations of FORMOSAT-3/COSMIC during 2007-2014	Y-I. Chen	
1155-1210	Characterization of GNSS amplitude scintillation occurrences over African and American Longitudes at varying elevation angles during the maximum phase of solar cycle 24	A. Akala	
1210-1225	ESA Monitor Measurement campaign results over West Africa	Y. Béniguel	
1225-1400	Lunch Break		
Session 5			
1400-1420	Short-term tidal variability in the mesosphere, thermosphere, and ionosphere. <i>(Invited)</i>	N. Pedatella	Immel & Obrou

1420-1435	Long term tidal variability as observed from Satellite and ground based observations	D. Singh	
1435-1450	Influence of DE3 tide on the equinoctial asymmetry of the zonal mean ionospheric electron density	Z. Ren	
1450-1505	Meridional Wind Gradients in Equatorial Thermospheric Dynamics	S. Sanders	
1505-1520	Geomagnetic conjugate observations of plasma bubbles and a brightness wave using airglow imagers and Fabry-Perot interferometers	K. Shiokawa	
1520-1535	"Double-humped" structure in latitudinal distribution of mesospheric temperatures revealed during sudden stratospheric warming events	R. P. Singh	
1535-1555	Thermosphere/Ionosphere responses to SSWs simulated by GAIA model. (<i>Invited</i>)	H. Liu	
1555-1620	Coffee Break		
1620-1640	Vertical coupling between the middle atmosphere and ionosphere: relative importance of different mechanisms. (<i>Invited</i>)	L. P. Goncharenko	
1640-1655	A theoretical study of the convective instability and subsequent generation of acoustic-gravity waves in the troposphere to MLT region	B. R. Tiwari	
			Gurubarn & Nava
1655-1710	Investigation Of Gravity Wave Activity As A Possible Mechanism Of Daytime Mstids Of The South-America Sector.	O.F. Jonah	
1710-1725	Decay of nighttime electrified medium-scale traveling ionospheric disturbances approaching lower latitudes	V. L. Narayanan	
1725-1740	Analysis of occurrence characteristics of sporadic E	H. Shinagawa	

	layers using GAIA		
1740-1755	A case study on occurrence of an unusual structure in the sodium layer over Gadanki, India	S. Sarkhel	
1755-1815	Results from Tidal and Planetary Wave Analysis of the Ionosphere using FORMOSAT-3/ COSMIC, TIMED, and GPS-TECs. (<i>Invited</i>)	L. C. Chang	
Friday (23 October 2015)			
Session 6			
0800-0830	F-region currents at low latitudes deduced from the Swarm satellite constellation. (<i>Invited</i>)	C. Stolle (H. Lühr)	
0830-0900	A high altitude balloon experiment to probe stratospheric electric fields from low latitudes. (<i>Invited</i>)	S. Gurubaran	Pallamraju & Yamamoto
0900-0930	Understanding Ionospheric Connections to Sun and Earth. (<i>Invited</i>)	T. J. Immel	
0930-1000	HF Radar for Long-Range Monitoring of Ionospheric Irregularities in the Equatorial Region. (<i>Invited</i>)	R. T. Parris	
1000-1030	Coffee Break		
1030-1045	New horizons in studies of the ionospheric plasma irregularities with sounding rockets, satellites, and numerical modeling	W. J. Miloch	
1045-1100	Development of realtime 3D tomography of the ionosphere from GPS-TEC in Japan	M. Yamamoto	Shiokawa & Heine
1100-1115	A future vision: Incoherent and coherent scatter radar in Ethiopia	A. J. Coster	
1115-1130	Global-scale Observations of the Limb and Disk	E. R. Talaat	

	(GOLD) Mission - Observing Forcing of the Thermosphere-Ionosphere System from Above and Below	(T. J. Immel)	
1130-1145	Measurement of Traveling Ionospheric Disturbances by Radio Methods Using Transmitters of Opportunity	J. LaBelle	
Panel Session			
1145-1300	Panel discussion about future plans		Erickson
	Adjourn		
Afternoon	Open for other side meetings		

Appendix C

List of Countries represented at the meeting

1. Brazil
2. Cameroon
3. China
4. Cote d'Ivoire
5. Czech Republic
6. Egypt
7. Ethiopia
8. France
9. Germany
10. India
11. Italy
12. Japan
13. Kenya
14. Malaysia
15. Morocco
16. Nigeria
17. Norway
18. Peru
19. Russia
20. Rwanda
21. South Africa
22. Sweden
23. Taiwan
24. Uganda
25. UK
26. Ukraine
27. USA

Appendix D

List of Participants and their contact information

Last Name	First Name	E-mail	Country
Adeniyi	Jacob O.	joadeniyi2k15@gmail.com	Nigeria
Admas	Frezer		Ethiopia
Admass	Fekadu		Ethiopia
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Coster	Anthea	ajc@haystack.mit.edu	USA
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Damtie	Shegaw	shegawd@yahoo.com	Ethiopia
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Getachew	Nigist		Ethiopia
Getahun	Balew		Ethiopia
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Appendix E

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